# Assignment 1: Logit / Probit – Income Inequality and Happiness

Advanced Methods for Data Analysis in Public Policy Research FS 26 —- ETH Zürich

2026-03-09

## Application Background

This assignment uses the application context of Alesina, Di Tella & MacCulloch (2004), who study how income inequality is related to individuals' self-reported happiness. The purpose of the assignment is to practice estimation and interpretation of binary outcome models: the Linear Probability Model (LPM), logit, and probit.

**Research Question**: How does income inequality affect the probability that an individual reports being happy?

## Instructions

- Provide all code and answers within the provided markdown template.
- Ensure that you formulate a written answer where asked. Be concise.
- **Pass requirements**:
  - The analysis must be replicable on any machine. Use relative file paths when loading the data.
  - Completion of all questions is mandatory for a passing grade (Estimated effort: 6 hours).
- **When you are done**: Submit the Markdown script **and** the compiled HTML file on Moodle.
- **Submission deadline**: 16.03.2026, 9:00 am.
- **Presentation**: The assignment will be discussed during the exercise session on 30.03.2026 Be prepared to present the assignment in class. Every group member must be able to discuss and defend any part of the assignment.

## Exercises

1. **Descriptives**
   a) Report summary statistics (mean, sd, min, max) for *happy*, *ln_income*, *gini*, *age*, *educ_yrs*.
   b) Report the share of *happy* in the full sample and by unemployment status.
   c) Briefly describe whether individuals in higher-inequality regions report lower happiness.
2. **Linear Probability Model**
   a) Estimate the LPM:
   $$happy_i = \alpha + \beta_1 ln_income_i + \beta_2 gini_i + \epsilon_i$$
   b) Interpret $\beta_1$ and $\beta_2$ as changes in probability.
   c) Do any fitted values fall outside $[0, 1]$? If yes, report the share and show an example.

d) In one paragraph, discuss limitations of the LPM for binary outcomes.

3. **Logit and Probit**

   a) Estimate logit and probit versions of the model in (2).

   b) Compare signs and statistical significance of *ln_income* and *gini* across models.

   c) Explain why logit/probit coefficients are not directly interpretable as percentage-point changes.

4. **Adding controls and clustering**

   a) Estimate a preferred specification (logit or probit) including controls: *age, age2, female, educ_yrs, married, unemployed*. Use standard errors clustered at the region level.

   b) Report the estimated effects of *ln_income* and *gini* using marginal effects.

   c) Compare to the bivariate model. How do the estimates change?

   d) Which control variables are most strongly associated with happiness?

5. **Predicted probabilities for profiles**

   a) Using your preferred nonlinear model from (4), compute predicted happiness probabilities for the following profiles:

      A age=30, female=0, educ=12, married=0, unemployed=0, *ln_income* at the 25th percentile, *gini* at the 25th percentile.

      B same as A but *gini* at the 75th percentile.

      C same as A but *ln_income* at the 75th percentile (*gini* at the 25th percentile).

   b) Interpret the differences ($B - A$ and $C - A$).

6. **Marginal effect of a 10% income increase**

   a) Report the average marginal effect (AME) of *ln_income*.

   b) Based on the AME, approximate the effect of a 10% income increase ($\Delta ln \approx 0.10$).

   c) Explain why marginal effects can vary across individuals and why AME is often reported.

7. **Interaction / heterogeneity**
   Alesina et al. discuss that inequality may affect groups differently

   a) Estimate a model that includes an interaction between *gini* and *income_z* (relative income). Interpret the interaction in terms of probabilities.

8. **Policy communication and limitations**

   a) Give two limitations of interpreting coefficient on inequality causally.

   b) Write a short policy-facing interpretation (4–6 sentences) of your main result using predicted probabilities or marginal effects.

# Appendix

## Data

**File**: `happiness_inequality_teaching.csv`

**Unit of Observation**: Individual

| Variable | Description |
|---|---|
| *happy* | 1 if the individual reports being happy |
| *ln_income* | *log*(income) |
| *income* | income in levels |
| *gini* | inequality measure at the region level (Gini coefficient) |
| *region_id* | numeric region identifier |
| *region* | region name |
| *age* | individuals' age |
| *age*2 | individuals' age squared |
| *female* | 1 if the individual is female |
| *educ_yrs* | years of education |
| *married* | 1 if married |
| *unemployed* | 1 if unemployed |
| *income_z* | relative income position (standardized within region) |

## References

Alesina, A., Di Tella, R., & MacCulloch, R. (2004). Inequality and Happiness: Are Europeans and Americans Different? Journal of Public Economics, 88(9–10), 2009–2042.